



MMADPE: drug repositioning based on multi-hop graph Mamba aggregation with dual-modality graph positional encoding

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Abstract

Drug repositioning (DR) has emerged as a critical research strategy for uncovering novel therapeutic applications of existing drugs, demonstrating considerable clinical significance. Despite promising advancements in computational methods for predicting drug–disease associations, most algorithms exhibit three major limitations. First, they inadequately capture high-order relationships within drug–disease networks. Second, they fail to concurrently model both local interaction strengths and global network topologies. Most importantly, current models lack biological interpretability and are incapable of extracting meaningful biological patterns from numerical data. To address these challenges, we propose a novel drug repositioning framework, MMADPE. This framework constructs a similarity network supporting multi-hop aggregation and leverages the linear complexity of Graph Mamba to efficiently integrate multi-order neighborhood information, thereby significantly enhancing the modeling of long-range drug–disease interactions. Subsequently, a dual-modal graph positional encoding is employed, capturing global network topology via Laplacian eigenvectors and characterizing local node association strengths through random walk statistics. Finally, the framework incorporates a GraphGPS hybrid architecture that fuses gated graph convolution with Transformer attention mechanisms to extract molecular biochemical features and their semantic relationships, achieving a deep integration of topological structures and biological semantics. Extensive experiments on three benchmark datasets demonstrate that MMADPE consistently outperforms state-of-the-art methods in drug repositioning tasks. Notably, case studies on two common diseases combined with molecular docking experiments not only validate the effectiveness of our approach but also provide novel mechanistic insights into MMADPE’s ability to identify previously unrecognized drug–disease associations.

Keywords Drug repositioning · Drug–disease associations · Graph Mamba · GraphGPS · Graph positional encoding

Introduction

Drug discovery has long been a critical bottleneck in the biomedical field [1]. In recent years, with advancements in life science technologies such as genomics, significant breakthroughs have been achieved in drug research and

development [2]. However, the average time from discovery to approval for innovative drugs remains approximately 10 years, and over 90% of candidate drugs fail in early clinical trials due to unpredictable toxic side effects [3, 4]. The cost of developing a single new drug has now soared to \$2.6 billion, with substantial resources consumed in screening and early evaluation of ineffective compounds [5]. Faced with high costs and failure rates, the industry urgently needs more efficient R&D strategies. Although traditional wet-lab experiments can accurately validate drug–disease interactions, their inefficiency and labor-intensive nature remain significant drawbacks [6]. The successful application of computational methods has opened new avenues for drug development [7], among which computational drug repositioning (DR) has emerged as a key breakthrough due to its efficiency and cost-effectiveness, primarily divided into machine learning [8–10] and deep learning approaches.

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Machine learning-based drug repositioning (DR) methods predict drug–disease associations by integrating various algorithmic techniques. These primarily include matrix factorization [11, 12], matrix completion [13, 14], support vector machines [15–17], logistic regression [18, 19], and random forests [20, 21], among other classical approaches. Representative applications include Yang et al. [16] innovatively proposed a multiple Laplacian regularized SVM, constructing a multi-graph model with five Laplacian graphs and enhancing sample relationship representation through an improved edge-weight calculation method. Gottlieb et al. [18] systematically demonstrated the advantages of logistic regression in biomedical data analysis, including its ability to adjust for confounding factors and provide interpretable odds ratios, while validating model stability using the Framingham Heart Study. Kitsiranuwat et al. [20] employed random forests, leveraging their dual randomness to effectively handle high-dimensional data, while assessing feature importance via out-of-bag error estimation. Despite their success in DR, machine learning methods still face three major limitations: First, they heavily rely on large-scale, high-quality training data. Second, the feature engineering process demands extensive manual intervention. Most critically, their reliance on low-order embedding features struggles to capture the complex, deep biological associations and high-order interactions between drugs and diseases. Specifically, multi-graph construction and random forest feature selection still require expert knowledge, while the linear assumptions of logistic regression limit its ability to model nonlinear biological relationships. These constraints significantly hinder the predictive performance of machine learning models in complex biological systems—precisely the challenges that next-generation deep learning approaches aim to overcome.

To address the limitations of traditional machine learning methods in DR, researchers have shifted focus to deep learning approaches that demonstrate superior feature learning capabilities. In contrast to conventional methods dependent on manual feature engineering, deep learning employs multi-level nonlinear transformations to automatically extract high-order features, thereby significantly enhancing predictive performance. The field has witnessed a series of groundbreaking research achievements: In deep representation learning, deepDR [22] pioneered the use of multimodal autoencoders to integrate heterogeneous data sources, achieving comprehensive mining of drug–disease features. HNet-DNN [23] constructed heterogeneous information networks and utilized deep neural networks to capture complex interaction characteristics between drugs and diseases. Regarding collaborative filtering techniques, DRWBNCF [24] innovatively designed weighted bipartite network models combined with collaborative filtering algorithms, substantially improving the extraction capability

for high-order association features. Graph neural network methods have also achieved significant breakthroughs in this domain, for instance, GCNGAT [25] aggregated neighborhood information through graph convolution operations while incorporating attention mechanisms to dynamically adjust node weights. HINGRL [26] and DDAGDL [27] established multi-source biomedical heterogeneous networks and employed graph representation learning techniques to thoroughly explore potential biological associations. Particularly noteworthy is the innovative contribution of AMDGT [28]. This model integrates multimodal similarity data and biochemical interaction networks through a dual-graph Transformer architecture, utilizing an improved attention mechanism to simultaneously model relationships between both homogeneous and heterogeneous nodes. The specially designed AttentionFusion module enables adaptive fusion of feature representations from different modalities, achieving new heights in prediction performance while maintaining biological plausibility. These deep learning approaches employ end-to-end learning paradigms, successfully overcoming the inherent limitations of traditional methods in nonlinear feature learning and manual intervention, thereby opening new pathways for DR research.

Despite the effectiveness of graph neural networks in biomedical analysis, their performance remains constrained by fundamental limitations: most shallow architectures only aggregate direct neighbor information, failing to model multi-hop associations between drugs and diseases, while increasing depth often leads to gradient vanishing; node positional encoding strategies are oversimplified, typically choosing between global Laplacian eigenvectors and local random walks, making it difficult to balance micro-interactions and macro-network; insufficient data integration, as heterogeneous networks often rely on single-modal data, ignoring cross-modal features, such as drug molecular structures and disease ontology semantics, thereby limiting the biological interpretability of node representations. These challenges call for more sophisticated architectural designs.

To address these challenges, we propose a novel computational drug repositioning method called MMADPE. First, we construct drug similarity networks and disease similarity networks based on drug multimodal similarity and disease multimodal similarity. Then, we employ a low-dimensional density embedding strategy to reduce the dimensionality of the heterogeneous network, obtaining more information-dense association representations. We introduce a graph Mamba with multi-hop neighbor aggregation capability to learn high-order interaction features from both the association representations and similarity networks, deriving similarity interaction features. Second, we encode the biochemical characteristics of drugs and diseases using mol2vec and mesh2vec. Based on these biochemical features, we perform both Laplacian positional encoding and

random walk positional encoding to simultaneously capture global topological structures and local connection probabilities. We then utilize graph GPS to mine their local and global interaction features, fusing them to obtain network interaction features. Finally, we perform parallel fusion of the similarity interaction features and network interaction features, employing a dot product decoder to predict potential drug–disease associations. The overall architecture of MMADPE is illustrated in Fig. 1. The main contributions of this study can be summarized as follows:

- Constructed as a similarity network and integrated into multi-hop neighborhood aggregation, it is capable of capturing high-order associations. Meanwhile, the linear complexity advantage of Graph Mamba is leveraged to efficiently integrate multi-order neighborhood information.
- Propose Dual-modality Graph Positional Encoding to comprehensively captures both global topological structures through Laplacian eigenvectors and local association strengths between nodes via random walks, thereby

generating more holistic node feature representations for drugs and diseases.

- Utilize a GraphGPS model constructed on a GateGCN-Transformer hybrid architecture to capture both local and global biochemical features of drug-disease interactions, generating node representations that integrate topological structures with biological semantic information.

The organization of this paper is as follows: Section 2 outlines the materials and methods employed in this research, detailing the dataset, data preprocessing steps, and specific information about the model. Section 3 presents the experimental results, including performance comparisons under various settings, parameter sensitivity analysis, extended studies on tripartite heterogeneous networks, ablation experiments, case studies, and molecular docking experiments. Finally, Section 4 concludes with a summary of our findings.

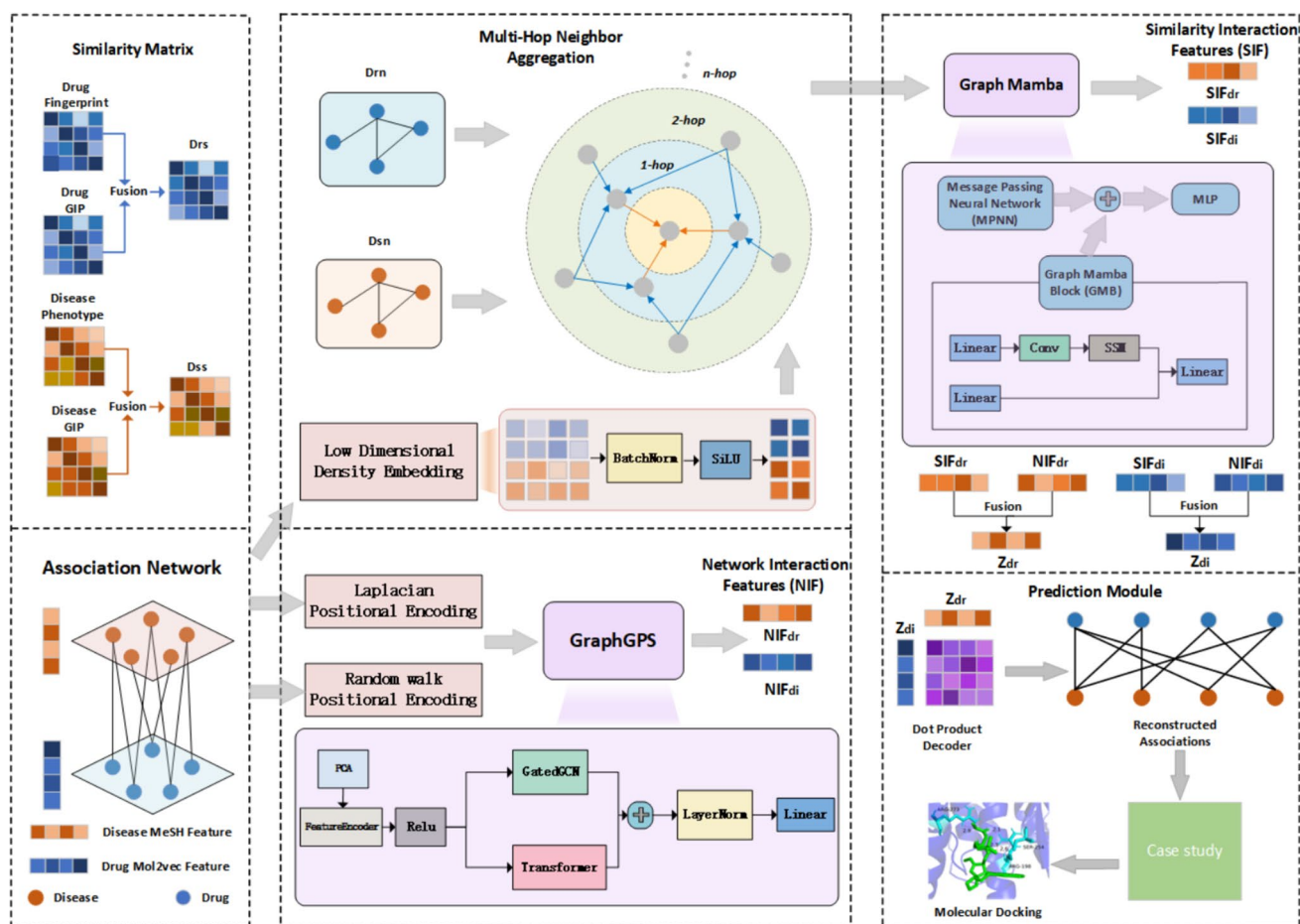


Fig. 1 Model flowchart of MMADPE

Materials and methods

Datasets

This study employs three benchmark datasets for performance evaluation. The B-dataset is derived from the Comparative Toxicogenomics Database (CTD) [29] by Zhang et al. [30] and encompasses 269 drugs, 598 diseases, and 18,416 validated drug–disease associations. The C-dataset, constructed by Luo et al. [31], contains 2,532 known drug–disease associations, covering 663 drugs and 409 diseases. The F-dataset, developed by Gottlieb et al. [19], comprises 1,933 high-confidence known drug–disease associations (DDA), including 593 validated drugs from the DrugBank database [32] and 313 diseases from the OMIM database [33]. These three datasets are widely adopted benchmarks in the field of drug repositioning, and their sparsity levels (11.44%, 0.93% and 1.04%) align with the sparse characteristics of real-world biomedical networks. In the present study, negative samples were generated through random pairing of drugs and diseases without known associations. The dataset was carefully balanced to ensure equal numbers of negative and positive sets. Table 1 summarizes the three datasets.

Similarity calculation

Drug fingerprint similarity

In this study, we adopted a method similar to that of Yang et al. [34] to evaluate the similarity between drugs by calculating the similarity of their molecular fingerprints. This process involves the Simplified Molecular Input Line Entry Specification (SMILES) of the drugs, using RDKit and the Tanimoto [35] method to obtain fingerprint similarity. Specifically, we first extract the molecular fingerprints of each drug and then compute the similarities between these fingerprints to form a matrix that represents the similarity between drugs. This matrix is denoted as $D_{rf} \in \mathbb{R}^{m \times m}$, where m is the total number of drugs. Through this approach, we can quantify the relationships between drugs, providing a foundation for subsequent analyses.

Table 1 Details of datasets

Dataset	Drug	Disease	Known association	Sparsity
B-dataset	269	598	18416	0.1144
C-dataset	663	409	2352	0.0093
F-dataset	593	313	1933	0.0104

Disease phenotype similarity

Phenotype similarity quantifies how similar diseases are based on text mining of medical information from the OMIM database. In this study, we specifically utilized Mim-Miner [36] to calculate the phenotype similarity scores. The resulting similarity is represented as a matrix $D_{sp} \in \mathbb{R}^{n \times n}$, where n denotes the total number of diseases. This quantitative representation facilitates downstream analyses by providing a principled measure of disease similarity at the phenotypic level.

Gaussian interaction profile (GIP) Kernel similarity of drugs and diseases

Due to the incomplete information and high sparsity of both drug fingerprint similarity and disease phenotype similarity, we use Gaussian Interaction Profile (GIP) kernel similarity to address this issue. It is hypothesized that drugs associated with the same disease typically exhibit similar characteristics. Therefore, we can use the following formula to describe the GIP kernel similarity between drugs r_i and r_j :

$$D_{rg}(r_i, r_j) = \exp(-\phi_r \|IP(r_i) - IP(r_j)\|^2) \quad , \quad (1)$$

$$\phi_r = \frac{\phi'_r}{\frac{1}{m} \sum_{k=1}^m \|IP(r_k)\|^2} \quad , \quad (2)$$

where $D_{rg} \in \mathbb{R}^{m \times m}$, and $IP(r_i)$ represents the corresponding column of drug r_i in the drug–disease association matrix, with ϕ'_r equal to 1. Similarly, the GIP kernel similarity of diseases is formulated as follows:

$$D_{sg}(s_i, s_j) = \exp(-\phi_s \|IP(s_i) - IP(s_j)\|^2) \quad , \quad (3)$$

$$\phi_s = \frac{\phi'_s}{\frac{1}{n} \sum_{k=1}^n \|IP(s_k)\|^2} \quad , \quad (4)$$

where $D_{sg} \in \mathbb{R}^{n \times n}$, and $IP(s_i)$ denotes the corresponding row of disease s_i , with ϕ'_s set to 1.

Similarity integrated

We obtained the similarities between drugs and diseases using several different methods and combined them with certain weighting coefficients to derive the final similarity matrix for drugs and diseases, calculated as follows:

$$D_{rs}(r_i, r_j) = \begin{cases} \frac{D_{rf}(r_i, r_j) + D_{rg}(r_i, r_j)}{2}, & \text{if } D_{rf}(r_i, r_j) \neq 0 \\ D_{rg}(r_i, r_j), & \text{otherwise} \end{cases} \quad , \quad (5)$$

$$D_{ss}(s_i, s_j) = \begin{cases} \frac{D_{sp}(s_i, s_j) + D_{sg}(s_i, s_j)}{2}, & \text{if } D_{sp}(s_i, s_j) \neq 0 \\ D_{sg}(s_i, s_j), & \text{otherwise} \end{cases} \quad (6)$$

The integrated drug and disease similarity matrices are represented as D_{rs} and D_{ss} , based on the similarity matrix, the corresponding similarity networks $D_{rm} \in \mathbb{R}^{m \times m}$ and $D_{sn} \in \mathbb{R}^{n \times n}$ are constructed by setting a threshold ϵ . The selection of this threshold is crucial. If the threshold is set too high, the network will become overly sparse, resulting in loss of effective information. If the threshold is set too low, the network will become overly dense, introducing excessive noise. Therefore, an appropriate threshold must be selected to obtain a similarity network with practical significance.

Initial biochemical feature

This study innovatively adopts pre-trained language model embedding technology to characterize node features. Compared with traditional methods, this approach demonstrates greater flexibility and robustness, yet this technical pathway has long been neglected in previous research. For drug nodes, the mol2vec method [37] is employed to generate 300-dimensional feature representations ($H_{dr} \in \mathbb{R}^{m \times 300}$), which can accurately characterize the chemical structural features of drug molecules at a high level. For disease nodes, we utilize the MeSHHeading2vec algorithm [38] to transform the MeSH tree structure [39] into 64-dimensional disease vector representations ($H_{di} \in \mathbb{R}^{n \times 64}$) that are rich in both biological and semantic information. However, since the second dimensions of the two biochemical features differ and cannot be computed simultaneously, we follow the approach of Liu et al. [40] and employ PCA [41] for noise reduction and dimensionality reduction, ultimately obtaining standardized 64-dimensional initial features ($H_{dr}^{(0)} \in \mathbb{R}^{m \times 64}$ and $H_{di}^{(0)} \in \mathbb{R}^{n \times 64}$). However, for a detailed process, please refer to Fig. 2, which contains a comprehensive explanation from this module to module 2.6.

Low-dimensional density embedding

To compress the sparse binary association matrix into low-dimensional dense vectors and to transform the discrete association relationships with values of 0 or 1 into distributed representations that capture topological structures and semantic associations, we design the following low-density embedding strategy, mathematically expressed as follows:

$$E_{dr} = \text{SiLU}(\text{BatchNorm}(AW_1 + b_1))W_2 + b_2 \quad (7)$$

$$E_{di} = \text{SiLU}(\text{BatchNorm}(A^T W_3 + b_3))W_4 + b_4 \quad (8)$$

where $A \in \{0, 1\}^{m \times n}$ denotes the binary drug–disease association matrix with m drugs and n diseases, in which a value of 1 indicates a known association and 0 represents an unknown relationship. The transpose of A is given by A^T . The matrices $E_{dr} \in \mathbb{R}^{m \times d}$ and $E_{di} \in \mathbb{R}^{n \times d}$ represent the output drug and disease embedding matrices, respectively, with d being the embedding dimension, set to 128. The weight matrices W_1, W_2, W_3 , and W_4 are learnable parameters, and b_1, b_2, b_3 , and b_4 denote the corresponding bias terms. SiLU refers to the self-gated activation function, and BatchNorm stands for the batch normalization operation.

Multi-hop neighborhood aggregation

The multi-hop neighborhood aggregation [42] models the cascade effects in complex systems, extending local associations to global systemic interactions. By leveraging multi-path cross-validation and signal attenuation mechanisms, it automatically reduces noise interference. Ultimately, through adjusting the hop range, we obtain drug–disease feature representations that encompass both molecular-level interpretability and global interactivity, computed as follows:

$$M_{dr} = [E_{dr}, \bar{D}_m E_{dr}, \bar{D}_m^2 E_{dr}, \dots, \bar{D}_m^k E_{dr}]^T \quad (9)$$

$$\bar{D}_m = (\text{diag}(\mathcal{D}_m \eta))^{-1/2} \mathcal{D}_m (\text{diag}(\mathcal{D}_m \eta))^{-1/2} \quad (10)$$

We use drugs as an example to illustrate the computational process, where $\bar{D}_m \in \mathbb{R}^{m \times m}$ denotes the normalized drug reachability graph, $D_m \in \mathbb{R}^{m \times m}$ represents the previously constructed drug similarity network, η is the normalization coefficient vector, and k indicates the number of hops. The operator $[\cdot]$ performs stacking along the hop-count dimension, yielding the multi-hop aggregated drug embedding matrix $M_{dr} \in \mathbb{R}^{(k+1) \times m \times 128}$, which is constructed by stacking drug embeddings from different propagation hops along the first dimension. Similarly, the disease embedding matrix $M_{di} \in \mathbb{R}^{(k+1) \times n \times 128}$ can be obtained through the identical operation.

Graph Mamba

For the requirement of capturing higher-order interaction features in drug-disease association prediction, traditional Graph Neural Networks (GNNs) [43] struggle to model global dependencies due to their over-reliance on local neighborhood aggregation. While Graph Transformer [44] can capture long-range relationships, their attention mechanism's $\mathcal{O}(N^2)$

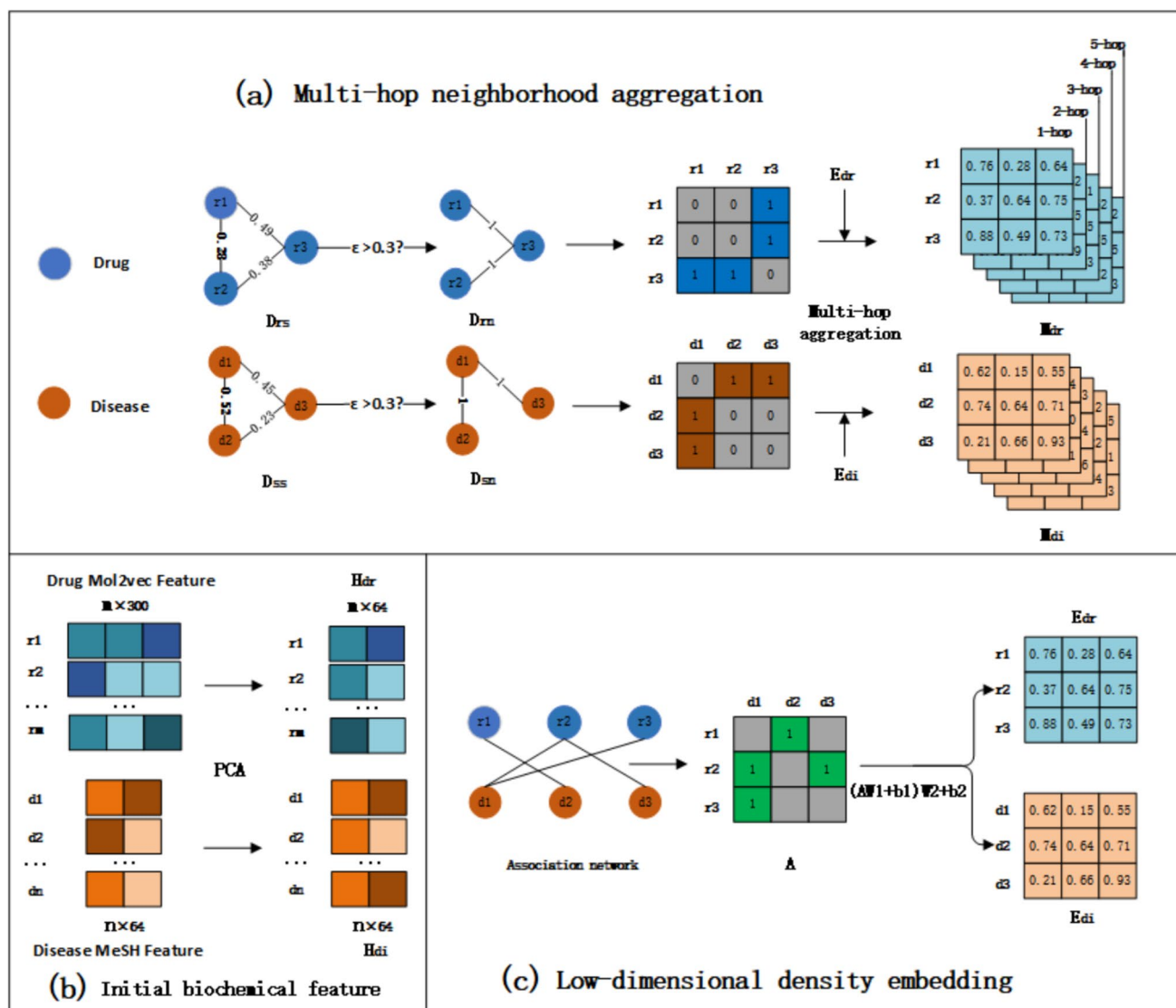


Fig. 2 A toy diagram of the process for initializing biometric features and multi-hop aggregation information in MMADPE.

computational complexity limits large-scale applications. To address these challenges, we introduce the Graph Mamba model [45].

Message passing neural network (MPNN)

The edge gating mechanism in the MPNN module can dynamically regulate information flow between nodes, effectively integrating node features M_{dr} and edge features E_{dr} in the drug–disease network. The method is described as follows:

$$M_{dr,i}^{(l+1)} = M_{dr,i}^{(l)} + \text{ReLU} \left(W_a M_{dr,i}^{(l)} + \sum_{j \in \mathcal{N}(i)} \eta_{ij} \odot W_b M_{dr,j}^{(l)} \right), \quad (11)$$

$$\eta_{ij} = \sigma \left(W_c M_{dr,i}^{(l)} + W_d M_{dr,j}^{(l)} + W_e E_{dr,ij}^{(l)} \right). \quad (12)$$

We continue to use drugs as an example, where $M_{dr,i}^{(l)}$ represents the feature vector of drug node i at layer l , $\mathcal{N}(i)$ denotes the set of neighboring nodes of node i , $\eta_{ij} \in [0, 1]$ represents the gated weight of edge (i, j) , σ denotes the Sigmoid activation function, $M_{dr,i}^{(l+1)}$ represents the updated feature of node i at layer $l + 1$, $\odot$ stands for the Hadamard product, and $E_{dr,ij}^{(l)}$ indicates the edge feature vector between drug nodes i and j .

After L layers of applying (11) and (12), we collect the final node features into

$$X_{\text{MPNN}} = [M_{dr,1}^{(L)}, M_{dr,2}^{(L)}, \dots, M_{dr,m}^{(L)}]^T \quad (13)$$

which serves as the definitive output of the MPNN branch.

State-space modeling (SSM)

The SSM module overcomes the locality limitation of traditional GNNs through global sequence modeling, enabling efficient capture of long-range dependencies. It arranges the input sequence in descending order of node degrees, allowing high-degree nodes to obtain broader contextual awareness, thereby extracting higher-order interaction features. The procedure is as follows:

$$\hat{X} = \text{LayerNorm}(P M_{dr}), \hat{Y} = \text{SSM}(\hat{X}), X_{\text{GMB}} = P^{-1} \hat{Y}, \quad (14)$$

where P is a permutation matrix constructed based on node degree (descending order), P^{-1} is its inverse, $\hat{X} = [x_1, x_2, \dots, x_m]^T$ is the sorted and normalized input sequence with x_t the feature vector of the t -th node, $\text{SSM}(\cdot)$ represents the state-space modeling operation whose output is $\hat{Y} = [y_1, y_2, \dots, y_m]^T$, and X_{GMB} is the final node representation restored to the original order.

The core update equations of the state-space model are as follows:

$$h_t = \bar{U}_t h_{t-1} + \bar{V}_t x_t, \quad y_t = C_t h_t, \quad (15)$$

where h_t is the hidden state at step t , h_{t-1} is the previous hidden state, $\bar{U}_t$ and $\bar{V}_t$ are the discretized state transition and input projection matrices, respectively, and C_t is the output projection matrix mapping h_t to y_t . The discretization operators are defined as follows:

$$\Delta_t = \text{softplus}(W_\Delta x_t), \quad \bar{U}_t = e^{\Delta_t U}, \quad \bar{V}_t = (\Delta_t U)^{-1} (e^{\Delta_t U} - I) \Delta_t V, \quad (16)$$

where Δ_t is a learnable step size obtained via a softplus on x_t , W_Δ is a trainable projection matrix, U and V denote the continuous-time state transition and input projection matrices, respectively, $\exp(\Delta_t U)$ is the matrix exponential, and I is the identity matrix.

$$\text{SIF}_{dr} = \text{MLP}(X_{\text{MPNN}} + X_{\text{GMB}}). \quad (17)$$

Finally, the Mamba module output X_{GMB} is fused with the node representations X_{MPNN} from the gated graph convolution branch, and the drug similarity interaction features SIF_{dr} are extracted through a multilayer perceptron. Similarly, the disease similarity interaction features SIF_{di} can be obtained in the same way.

Laplacian positional encoding

Laplacian positional encoding [46] is generated from the eigenvectors of the graph Laplacian matrix to encode the global topological structure of molecules. To rigorously define

spectral positional encoding on the heterogeneous drug–disease network, we first embed the bipartite association matrix $A \in \mathbb{R}^{m \times n}$ into a symmetric bipartite adjacency matrix:

$$\tilde{A} = \begin{bmatrix} \mathbf{0}_{m \times m} & A \\ A^T & \mathbf{0}_{n \times n} \end{bmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}, \quad (18)$$

where $\mathbf{0}_{m \times m}$ and $\mathbf{0}_{n \times n}$ are zero matrices ensuring $\tilde{A}$ is symmetric. Let $D = \text{diag}(\tilde{A}\mathbf{1})$ be the degree matrix representing the degrees of nodes, where $\mathbf{1}$ is the all-ones vector. The unnormalized graph Laplacian matrix is defined as follows:

$$T = D - \tilde{A}. \quad (19)$$

We then compute the first α non-trivial eigenpairs:

$$T\Phi = \Phi\Lambda, \quad (20)$$

where $\Phi = [\phi_1, \dots, \phi_\alpha] \in \mathbb{R}^{(m+n) \times \alpha}$ is the matrix composed of α eigenvectors, each column is an eigenvector. $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_\alpha) \in \mathbb{R}^{\alpha \times \alpha}$ is the diagonal eigenvalue matrix.

Due to sign ambiguity of eigenvectors, we unify the sign by the following standard alignment to ensure consistent sign orientation for each eigenvector:

$$s_i = \text{sign}\left(\sum_v \phi_{v,i}\right), \quad \phi_i \leftarrow s_i \phi_i. \quad (21)$$

The matrix Φ captures each node's structural role, representing the node's position in the global graph topology, with spectral coordinates approximating diffusion or resistance distances.

To endow these spectral coordinates with a contextual information across eigenmodes (semantic role), we build a compact mode-level token for each eigenmode i :

$$\mathbf{x}_i = [\lambda_i, \mu_i, \sigma_i] \quad (22)$$

$$\mu_i = \frac{1}{m+n} \sum_v \phi_{v,i}, \quad \sigma_i = \sqrt{\frac{1}{m+n} \sum_v (\phi_{v,i} - \mu_i)^2}, \quad (23)$$

where μ_i and σ_i represent the mean and standard deviation of the eigenvector entries, respectively, summarizing its distribution. This forms $X_{\text{mode}} \in \mathbb{R}^{\alpha \times 3}$.

A multi-head self-attention mechanism is then applied to contextualize inter-mode dependencies and learn mode importance weights $\mathbf{a} \in \mathbb{R}^\alpha$, which are finally used to mix the eigenvectors into node-wise encodings:

$$X = \sigma(W_\alpha X_{\text{mode}}), \quad Y = \text{SelfAttn}(X), \quad \mathbf{a} = \text{softmax}(Y \mathbf{w}_a), \quad (24)$$

$$P_{\text{LPE}} = \Phi \text{diag}(\mathbf{a}) W_L, \quad (25)$$

where W_α is the linear mapping matrix projecting node features to a higher dimension, $\sigma(\cdot)$ denotes the ReLU activation function, SelfAttn($\cdot$) is a multi-head self-attention encoder capturing interactions among nodes, w_a is a learnable vector generating attention scores, softmax($\cdot$) normalizes the weights $\mathbf{a}$, W_L maps spectral coordinates to the final positional encoding dimension, and P_{LPE} is the final Laplacian positional encoding matrix with each row corresponding to a node's encoding.

Random walk positional encoding

Random walk positional encoding [47] targets the local structural role, capturing node connectivity tightness and short-range reachability within neighborhoods. Using the same symmetric bipartite adjacency $\tilde{A}$, we define the row-stochastic transition matrix:

$$P = D^{-1}\tilde{A}, \quad (26)$$

where D is the degree matrix and P represents the transition probabilities from a node to its neighbors.

For each node i , we construct a vector of return probabilities at different walk lengths $z = 1, \dots, Z$:

$$RWPE_i = [(P^1)_{ii}, (P^2)_{ii}, \dots, (P^Z)_{ii}] \in \mathbb{R}^Z, \quad (27)$$

where $(P^z)_{ii}$ denotes the probability that a random walk starting at node i returns to node i at step z , quantifying the node's local connectivity tightness and short-range structural characteristics.

To inject a semantic role at the node level, a lightweight residual message passing mechanism is applied to contextually calibrate $RWPE_i$, yielding the final node-wise encoding:

$$P_{RWPE} = \text{MLP}_{\text{pos}}(RWPE_i) + \sum_{\theta=1}^{\Theta} \text{MsgPass}_{\text{pos}}(\{RWPE_j^{(\theta)}\}_{j \in \mathcal{N}(i)}), \quad (28)$$

where $\text{MLP}_{\text{pos}}(\cdot)$ is a multilayer perceptron for initial mapping of return probabilities, $\text{MsgPass}_{\text{pos}}(\cdot)$ is a residual message passing module aggregating neighbors' encodings, $\mathcal{N}(i)$ denotes the neighborhood of node i , and Θ is the number of message passing iterations. This approach combines the structural role from return probabilities with the semantic role from neighborhood-aware calibration, producing node-wise P_{RWPE} aligned row-wise with P_{LPE} to enable effective fusion in GraphGPS.

GraphGPS

GraphGPS [48] is a hybrid architecture that combines local graph convolution with global self-attention mechanisms, designed to model biochemical features and topological semantics in drug–disease interactions across multiple levels.

This hybrid architecture contains two fundamental components, namely a Feature Encoding and Extraction module and a Feature Fusion module.

Feature encoding and extraction

This module initially constructs the node representations $H^{(0)}$ by integrating the raw biochemical features with both Laplacian positional encodings and random walk positional encodings, thereby establishing a comprehensive foundation for downstream feature extraction. Subsequently, these representations are processed through two complementary pathways. In the local processing branch, GateGCN is employed to aggregate neighborhood information and extract structural features. In parallel, the global modeling branch leverages self-attention mechanisms to capture long-range dependencies and semantic patterns. This dual-branch architecture enables the model to simultaneously characterize local atomic interactions and global molecular relationships. The complete mathematical implementation is described below.

$$H^{(0)} = W_n H + W_{LPE} P_{LPE} + W_{RWPE} P_{RWPE}, \quad (29)$$

$$h_{\text{local}} = \text{GatedGCN}(H^{(0)}, E), \quad (30)$$

$$h_{\text{global}} = \text{Softmax}\left(\frac{(H^{(0)} W_Q)(H^{(0)} W_K)^T}{\sqrt{d_k}}\right)(H^{(0)} W_V), \quad (31)$$

where H represents the initially constructed biochemical features, which can denote either H_{dr} or H_{di} . P_{LPE} denotes the Laplacian positional encoding matrix, while P_{RWPE} represents the random walk structural encoding matrix. W_n , W_{LPE} , and W_{RWPE} are the linear projection matrices, yielding $H^{(0)}$ as the initial feature matrix. E stands for the edge feature matrix in the similarity network. W_Q , W_K , and W_V correspond to the query, key, and value projection matrices, respectively, with d_k representing the dimension of each attention head.

Feature fusion

The module integrates local and global features through residual connections to generate unified node representations. The mathematical implementation is as follows:

$$h = h_{\text{local}} + h_{\text{global}}, \quad (32)$$

$$\text{FFN}(h) = W_x \cdot \text{ReLU}(W_y \cdot h), \quad (33)$$

$$\text{NIF} = h + \text{FFN}(h). \quad (34)$$

We first combine the local features h_{local} with global features h_{global} through element-wise addition to obtain h ,

which preserves the complementary information from both features. Subsequently, a Feed-Forward Network (FFN) is employed to perform nonlinear transformations, enhancing the modeling capability of high-order feature interactions. The FFN structure includes dimension expansion and compression operations. Finally, Layer Normalization is applied

to stabilize the feature distribution, outputting the network interaction representation NIF.

Algorithm 1 Training procedure of MMADPE

Require: Drug reachability graph $\bar{D}_{rn}$ and disease reachability graph $\bar{D}_{sn}$; low-dimensional embedding E ; association network A ; drug and disease biochemical features H ; edge weight η ; number of epochs N .
 Ensure: Predicted probability matrix y' .

- 1: **for** fold = 1 **to** 10 **do**
- 2: **for** epoch = 1 **to** N **do**
- 3: $M_{dr} \leftarrow \text{MultiHop}(\bar{D}_{rn}, E_{dr})$
- 4: $M_{di} \leftarrow \text{MultiHop}(\bar{D}_{sn}, E_{di})$
- 5: $\text{SIF} \leftarrow \text{GraphMamba}(M, \eta)$
- 6: $P_{LPE} \leftarrow \text{LPE}(A)$
- 7: $P_{RWPE} \leftarrow \text{RWPE}(A)$
- 8: $\text{NIF} \leftarrow \text{GraphGPS}(H, P_{LPE}, P_{RWPE})$
- 9: $Z \leftarrow 1/2(\text{SIF} + \text{NIF})$
- 10: $y' \leftarrow \text{DotProductDecoder}(Z)$
- 11: $L \leftarrow \text{WeightedCrossEntropy}(y, y')$
- 12: Update model parameters to minimize L
- 13: **end for**
- 14: Compute validation metrics for this fold
- 15: **end for**
- 16: The average of the ten fold.

Prediction module

After completing the aforementioned steps, we fuse the similarity interaction features (SIF) output by Graph Mamba with the network interaction features (NIF) output by Graph GPS to obtain the final embedding representations Z . We employ a dot product decoder to obtain prediction scores for all edges, which correspond to the predicted scores for each drug-disease pair. Scores greater than or equal to 0.5 are considered as having an association, while those less than 0.5 are considered as having no association. The optimization is ultimately performed using the following weighted cross-entropy loss function:

$$\gamma' = \frac{1}{N} \sum_{(i,j) \in \mathcal{Y}^+ \cup \mathcal{Y}^-} [w_p y_{ij} \gamma + w_n (1 - y_{ij}) \gamma], \quad (35)$$

$$\gamma = - \sum_{(i,j) \in \mathcal{Y}^+ \cup \mathcal{Y}^-} \left[y_{ij} \ln(y'_{ij}) + (1 - y_{ij}) \ln(1 - y'_{ij}) \right], \quad (36)$$

where $\mathcal{Y}^+$ and $\mathcal{Y}^-$ denote the positive and negative sample collections, respectively, N is defined as the total number of samples. In this context, a specific drug i and disease j are denoted by the pair (i, j) . y_{ij} and y'_{ij} represent the true labels and predicted scores, respectively. w_p and w_n denote the weights of positive and negative samples.

The training procedure of the MMADPE model is illustrated in Algorithm 1. The model adopts 10-fold cross-validation, where each fold undergoes N training iterations. Specifically, the disease similarity network reachability graph $\bar{D}_{rn}$ and the drug similarity network reachability graph $\bar{D}_{sn}$ are first combined with low-dimensional embeddings E to perform multi-hop neighborhood aggregation, yielding the drug embedding matrix M_{dr} and the disease embedding matrix M_{di} . These matrices are then fed into the Graph Mamba network to generate the similarity interaction features (SIF).

Subsequently, Laplacian positional encoding and random walk positional encoding are simultaneously applied to the heterogeneous association network A to obtain features P_{LPE} and P_{RWPE} . These are combined with the drug biochemical

features H_{dr} and disease biochemical features H_{di} and jointly input into the Graph GPS network to obtain the network interaction features (NIF). Finally, the SIF and NIF are fused in parallel to generate the final embedding representation Z , which is then passed through a dot product decoder to produce the predicted probability matrix y' .

The model parameters are optimized in each iteration to minimize the loss, and the overall performance is evaluated based on the average metrics across the 10 folds.

Results and discussion

Experimental setup and evaluation metrics

In this research, the MMADPE model was executed using Python 3.12 and Pytorch 2.3.0 on an Nvidia RTX 4090 GPU. The training process utilized the Adam optimizer [49], with a learning rate of 0.0005 and a total training period of 500 epochs. During the experiments, the ratio of negative drug–disease pairs to positive samples was maintained at 1 to 1. The number of hops is set to 5, the similarity threshold is set to 0.3, the number of Graph Mamba layers is set to 1, and the number of attention heads in the Graph GPS is set to 8.

We used six standard evaluation metrics to assess the model's performance, including AUROC, AUPR, accuracy, precision, recall, and F1-score. By constructing the ROC curve, we elucidated the interplay between the model's sensitivity and specificity across varying thresholds, while the pre–recall curve elucidated the correlation between predictive accuracy and recall.

Parameter sensitivity analysis and performance evaluation

We investigated the influence of key hyperparameters in MMADPE: (1) The number of hops for multi-hop neighborhood aggregation, denoted as k . (2) The construction threshold for the similarity network, denoted as ϵ . (3) The number of layers in the Graph Mamba network, denoted as l_{mamba} . (4) The number of attention heads in Graph GPS, denoted as h . We conducted parameter analysis experiments on three datasets under 10-fold cross-validation (10-CV). Performing parameter sensitivity analysis is conducive to determining the optimal parameter combination and enhancing overall performance. The number of hops for multi-hop neighborhood aggregation is selected from the set $\{1, 2, 3, 4, 5, 6\}$, the similarity network threshold is adjusted within the range of $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, while the number of layers in the Graph Mamba network and the number of attention heads in Graph GPS are chosen from $\{1, 2, 3, 4, 5\}$ and $\{1, 2, 4, 8, 16\}$,

respectively. As shown in Fig. 3, the results of parameter analysis on three datasets are largely consistent.

Figure 3a indicates that the model attains peak performance when the neighborhood aggregation hop count is set to five, since very small hop counts capture only local neighborhoods and fail to model higher-order associations inherent to complex biological systems, whereas overly large hop counts can amplify spurious associations between unrelated nodes. Figure 3b shows that the drug–disease similarity network attains an optimal topology at $\epsilon = 0.3$ for the C-dataset and F-dataset, while the B-dataset reaches its best performance at $\epsilon = 0.5$; given the negligible performance difference between $\epsilon = 0.3$ and $\epsilon = 0.5$, we adopt $\epsilon = 0.3$ as the preferred threshold. Figure 3c demonstrates that a single Graph Mamba layer yields the strongest generalization, which can be attributed to Graph Mamba's combination of SSM-enabled global modeling and MPNN's dynamic gating mechanism that together realize multilayer representational capacity within a shallow architecture, thereby avoiding deep-network issues, such as over-smoothing and reduced computational efficiency. Figure 3d reveals that using eight attention heads achieves the best balance between biochemical feature disentanglement and computational efficiency, with each head specializing in particular classes of biochemical interactions and subspace orthogonality constraints mitigating informational redundancy.

Performance comparison under 10-fold cross-validation (10-CV) experiments

A 10-fold cross-validation (10-CV) was performed on the three benchmark datasets. Initially, the dataset was randomly divided into ten segments. For each fold of the experiment, one segment was designated as the test set, while the remaining nine segments were utilized for training. We compared the performance of MMADPE against seven state-of-the-art methods: deepDR [22], HNet-DNN [23], DRWBNCf [24], GCNGAT [25], HINGRL [26], DDAGDL [27], and AMDGT [28].

- DeepDR employs a multimodal deep autoencoder in conjunction with a variational autoencoder to extract features from diverse networks.
- HNet-DNN utilizes a deep neural network to forecast associations between drugs and diseases by extracting features from the heterogeneous drug–disease network.
- DRWBNCf develops a unique weighted bilinear graph convolution mechanism that leverages the neighborhood interaction neural collaborative filtering model to anticipate drug–disease associations.
- HINGRL constructs a heterogeneous information network using biological data and leverages graph representation learning methods to extract both topological and biological characteristics.

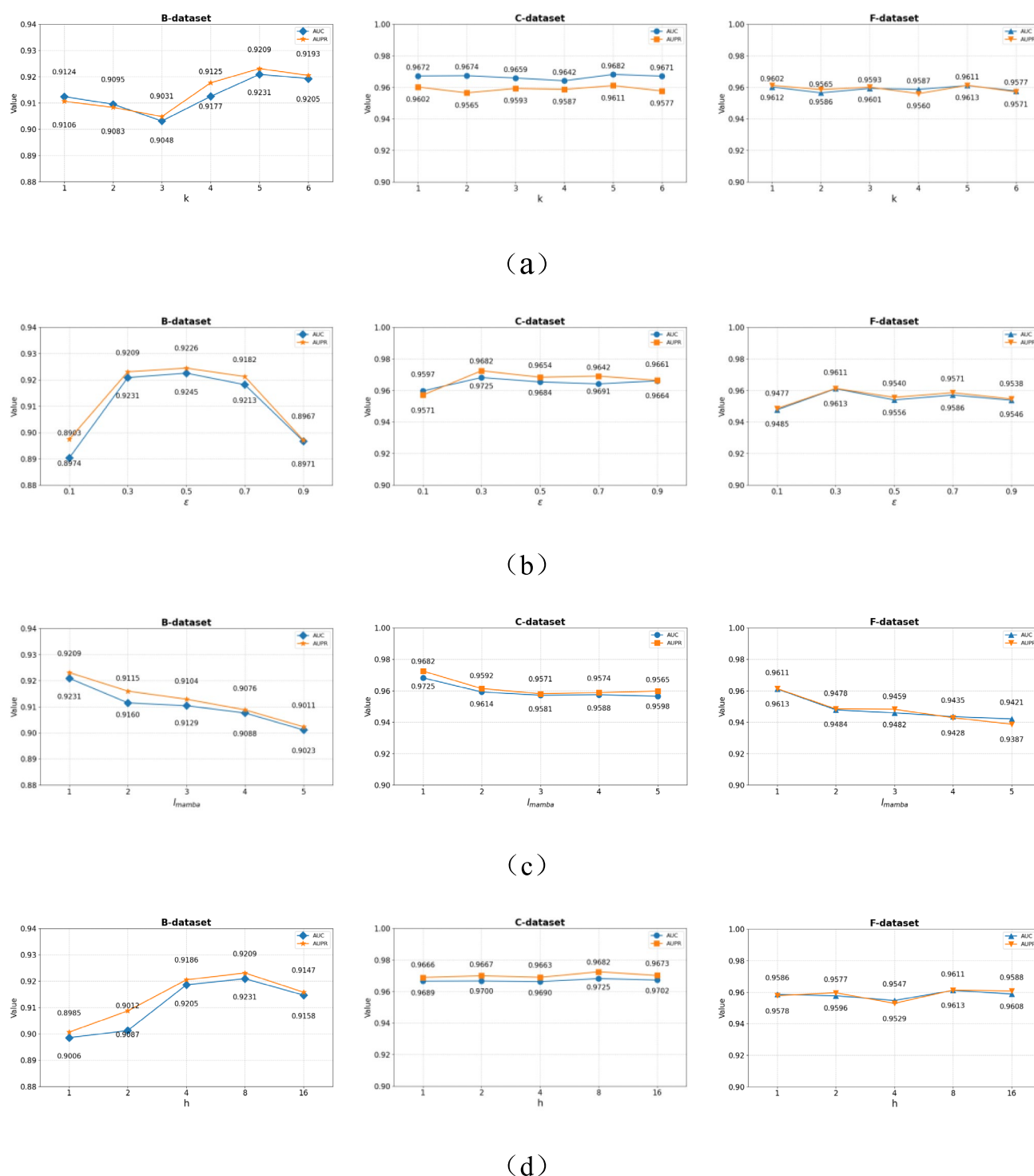


Fig. 3 Parameter sensitivity analysis (a) Parameter analysis of the number of multi-hop neighborhood aggregation hops k . (b) Parameter analysis of the similarity threshold ϵ . (c) Parameter analysis of

the number of network layers l_{mamba} in the MAMBA model. (d) Parameter analysis of the number of attention heads h in the GPS model

- GCNGAT designs a model that combines Graph Neural Networks with Global Average Pooling (GnnAp) to predict potential drug–disease associations.
- DDAGDL employs a geometric deep learning approach within a heterogeneous network, successfully capturing smoothed features by leveraging an attention mechanism.

Table 2 Performance comparison between MMADPE and benchmark methods across B-dataset, C-dataset, and F-dataset

Dataset	Model	AUC	AUPR	Accuracy	Precision	Recall	F1-score
B-dataset	deepDR	0.8172	0.8021	0.5994	0.8802	0.2334	0.3692
	GCNGAT	0.9092	0.9106	0.8268	0.8678	0.7711	0.8166
	HNet-DNN	0.8916	0.8908	0.8090	0.7814	0.8270	0.8036
	HINGRL	0.8834	0.8763	0.8024	0.8002	0.8073	0.8034
	DRWBNCf	0.9015	0.9003	0.5981	0.9821	0.2011	0.3364
	DDAGDL	0.8418	0.8304	0.7635	0.7605	0.7692	0.7648
	AMDGT	0.9173	0.9129	0.8396	0.8418	0.8425	0.8480
	MMADPE	0.9209	0.9231	0.8597	0.8533	0.9124	0.8976
C-dataset	deepDR	0.8893	0.9193	0.7384	0.9873	0.4543	0.6056
	GCNGAT	0.9417	0.9488	0.8712	0.8745	0.8673	0.8701
	HNet-DNN	0.9321	0.9466	0.8722	0.8919	0.8783	0.8799
	HINGRL	0.9372	0.9457	0.8541	0.8851	0.8533	0.8762
	DRWBNCf	0.9256	0.9329	0.8663	0.8984	0.8370	0.8612
	DDAGDL	0.9093	0.9137	0.8368	0.8074	0.8031	0.8007
	AMDGT	0.9598	0.9620	0.8957	0.8838	0.9115	0.8973
	MMADPE	0.9682	0.9725	0.9160	0.8975	0.9436	0.9176
F-dataset	deepDR	0.8645	0.8901	0.7602	0.9599	0.5241	0.6871
	GCNGAT	0.9182	0.9271	0.8463	0.8543	0.8360	0.8444
	HNet-DNN	0.9208	0.9168	0.8543	0.9285	0.8431	0.8413
	HINGRL	0.9306	0.9411	0.8641	0.8768	0.8444	0.8762
	DRWBNCf	0.9029	0.9183	0.8324	0.8784	0.8180	0.8315
	DDAGDL	0.9199	0.9285	0.8566	0.8505	0.8471	0.8496
	AMDGT	0.9477	0.9488	0.8805	0.8653	0.9032	0.8831
	MMADPE	0.9611	0.9613	0.9094	0.8903	0.9391	0.9078

Bold identifies the maximum value within each column

- AMDGT is a novel computational method based on multimodal learning and dual-graph Transformer for predicting drug–disease associations (see Table 2).

MMADPE demonstrates comprehensive advantages across all evaluation metrics for drug–disease association prediction. Compared to baseline methods, the model only slightly underperforms the optimal values in a few isolated metrics, confirming its superior generalization capability while maintaining prediction accuracy, with overall performance surpassing state-of-the-art methods in the field of DDA prediction (See Fig. 4).

In terms of robustness, MMADPE exhibits significant strengths. Taking the C-dataset as an example, deepDR achieves significantly higher AUC and AUPR than accuracy and F1-score, indicating that traditional methods are sensitive to the sample distribution. MMADPE addresses this issue through a low-dimensional density embedding strategy, which transforms binary association matrices into continuous vectors with topological semantics, effectively mitigating noise interference caused by data sparsity. Compared to GCN-based methods, such as GCNGAT and DRWBNCf, MMADPE achieves AUC improvements

of 2.83% and 4.60%, respectively. This enhancement can be attributed to its hybrid GraphGPS architecture, where the GateGCN branch extracts local biochemical features via edge gating mechanisms and the global attention branch captures long-range intermolecular interactions. These features are subsequently fused using residual connections to generate highly discriminative and interpretable NIF representations. When benchmarked against HNet-DNN, MMADPE delivers significant gains of 3.87% in AUC and 2.73% in AUPR. Although both approaches employ deep architectures, MMADPE’s feature encoding layer integrates raw biological features with DualPE positional encoding, thereby overcoming the limitation of traditional methods that neglect topological prior knowledge.

MMADPE’s innovative architecture also excels in computational efficiency. Specifically, compared to AMDGT, which adopts a multimodal dual-graph transformer, MMADPE improves AUC and AUPR by 0.81% and 1.12%, respectively. This advantage arises from the synergistic effect of its multi-hop neighborhood aggregation mechanism and Graph Mamba architecture, where

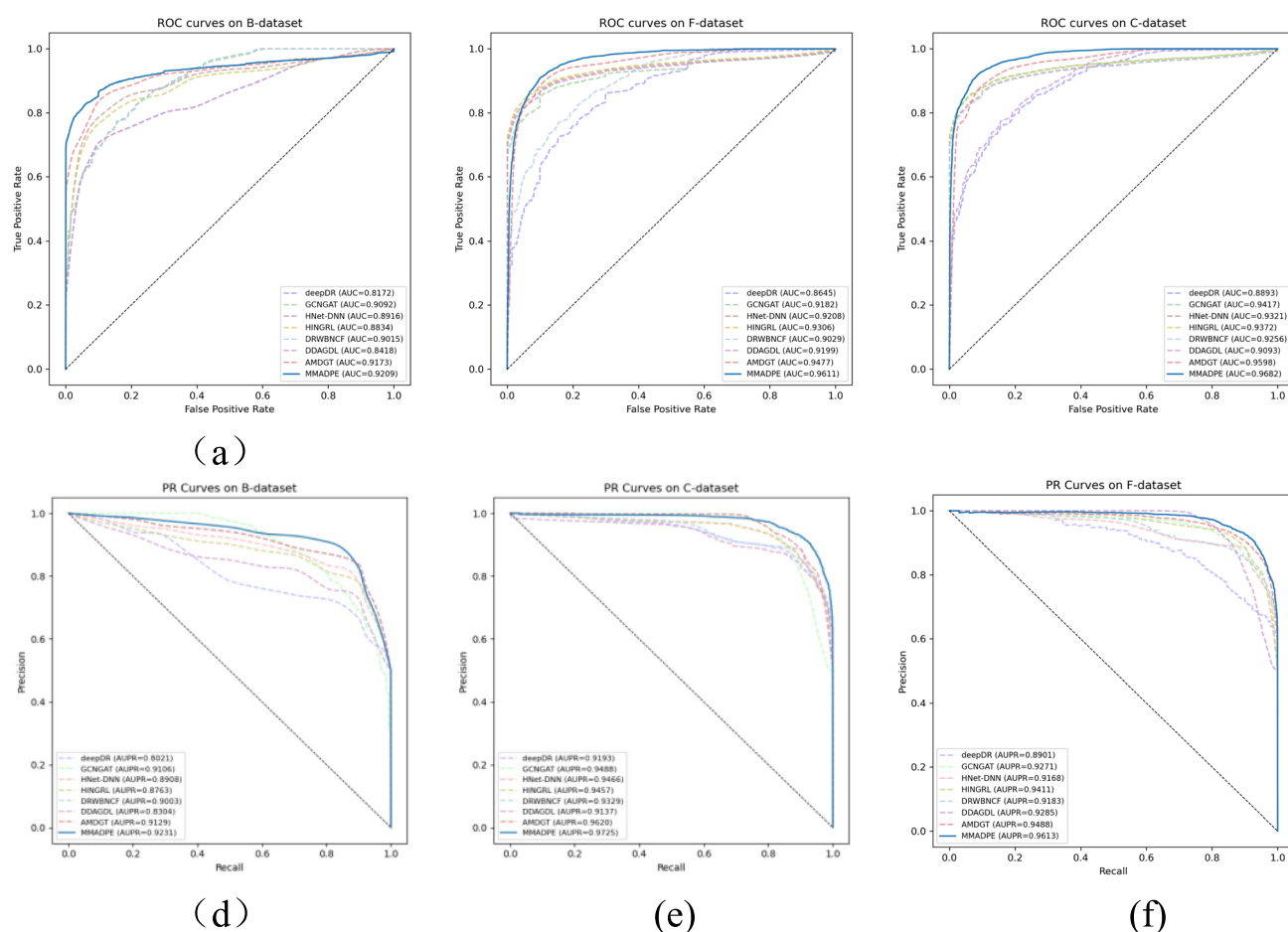


Fig. 4 The ROC and PR curves of different models for the benchmark datasets in the 10-CV experiment

Table 3 Comparative Analysis results on C-dataset

Model	AUC	AUPR	Accuracy	Precision	Recall	F1-score
GAE	0.8873	0.8744	0.8012	0.8162	0.8541	0.8299
VGAE	0.9005	0.8856	0.8193	0.8304	0.8693	0.8308
GT	0.8589	0.8458	0.7572	0.7221	0.8461	0.7732
HAN	0.9466	0.9531	0.8610	0.8874	0.8432	0.8724
GNN-DDI	0.9417	0.9410	0.8657	0.8961	0.8282	0.8602
MMADPE	0.9682	0.9725	0.9140	0.8975	0.9436	0.9176

Bold identifies the maximum value within each column

higher-order propagation matrices model molecular cascade effects while state-space modules capture long-range dependencies. Furthermore, MMADPE achieves a 3.53% higher F1-score than AMDGT, validating the effectiveness of Dual Positional Encoding in balancing precision and recall. Unlike AMDGT's attention mechanism, the Graph Mamba module employs a parameterized discretization approach, reducing computational complexity to linear levels while preserving global modeling

capabilities, thereby significantly accelerating training. This efficiency advantage, combined with hierarchical feature fusion, enables MMADPE to handle biomedical networks with millions of nodes, offering a novel technical pathway for large-scale drug repositioning studies.



Fig. 5 Performance comparison between Models on C-dataset

Table 4 Details of the Extended C-dataset

Dataset	Entities			Associations			Sparsity
	Drugs	Diseases	Proteins	Drug–disease	Drug–protein	Disease–protein	
C-dataset	663	409	993	2,532	3,773	10,734	0.0093

Comparative analysis with graph generative models and heterogeneous graph baselines

To more comprehensively evaluate the advantages of MMADPE as a biomedical link prediction model, we conducted comparative experiments against five baseline models on the C-dataset, including the graph generation-based foundational models GAE and VGAE [50], the widely used heterogeneous graph neural networks GT [51] and HAN [52], and the classic baseline model GNN-DDI [53]. As illustrated in Table 3 and Fig. 5, MMADPE consistently outperformed all five comparative models across every evaluation metric.

In capturing the complex, high-order biomedical patterns essential for drug–disease association prediction, MMADPE demonstrates superior capability compared to the graph reconstruction-based autoencoder approaches GAE and VGAE, owing to its innovative multi-hop graph Mamba aggregation and dual-modal graph positional encoding strategies. While the GT possesses global modeling capacity, its reliance on standard self-attention

mechanisms and lack of tailored encoding for the unique topological characteristics of biomedical networks limit its performance in drug–disease association tasks. Notably, even when compared to specialized heterogeneous graph models such as HAN and knowledge graph-enhanced approaches like GNN-DDI, MMADPE maintains strong competitiveness, achieving significant improvements in Accuracy, Recall, and F1-Score. These results substantiate that our model architecture generates more discriminative representations of drugs and diseases than existing heterogeneous graph models and domain-specific approaches, thereby attaining superior performance in biomedical link prediction.

Extension experiments on tripartite heterogeneous graph

To validate the generalization ability of MMADPE on richer multimodal biomedical graphs, we introduced proteins as mediating entities based on the existing drug–disease

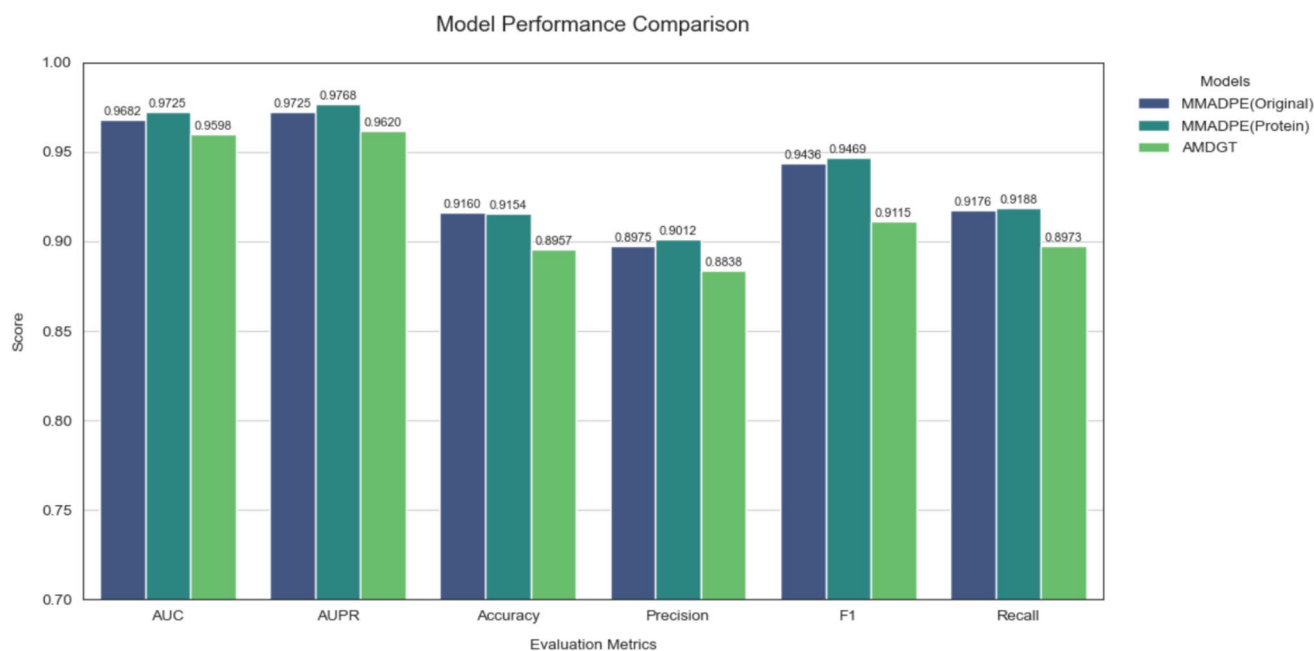


Fig. 6 Performance comparison between Models on C-dataset

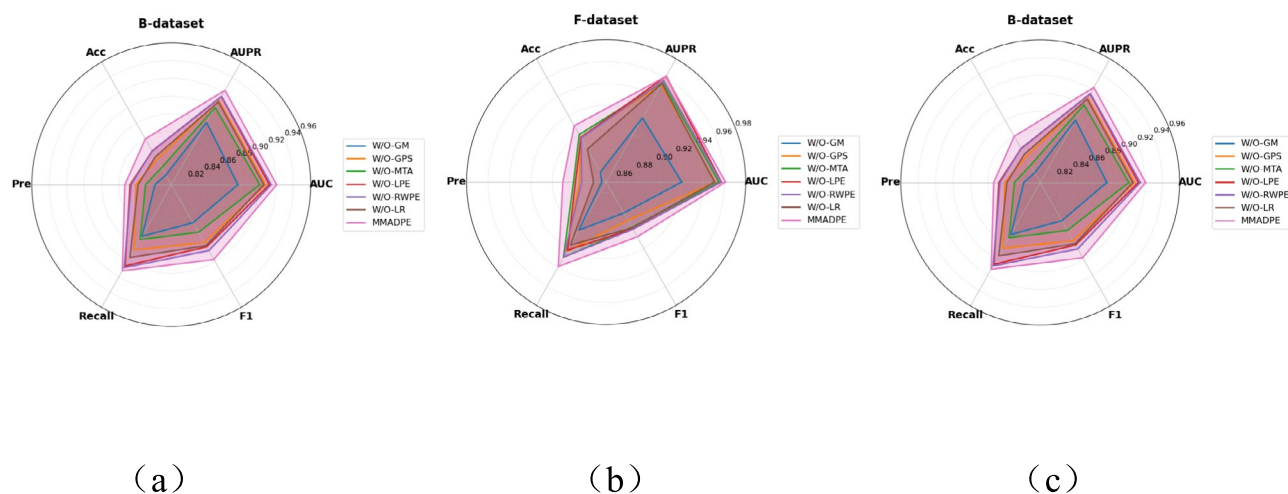


Fig. 7 Performance comparison between MMADPE and its variants on C-dataset and F-dataset

network of the C-dataset, constructing a drug–protein–disease tripartite heterogeneous graph. Table 4 summarizes the information from the extended C-dataset, where drug–protein interaction data is sourced from DrugBank and protein–disease associations from DisGeNET [54]. In this experiment, protein nodes are used as mediating nodes in the heterogeneous graph, with their initial biochemical features generated by the protein language model ESM-2 [55] and reduced to 64 dimensions through PCA to align with drug and disease features. The constructed heterogeneous

adjacency matrix A contains three types of bipartite edges: drug–disease, drug–protein, and protein–disease. Dual-modality Graph Positional Encoding generates positional features on this heterogeneous adjacency matrix to capture global and local topological information, followed by retaining the original process.

The experimental results are shown in Fig. 6 After incorporating protein mediating information, the overall performance of MMADPE significantly improved: AUC increased from 0.9682 to 0.9725, and AUPR increased by 0.43%,

Table 5 Performance comparison between MMADPE and its variants across C-dataset, F-dataset, and B-dataset

Dataset	Model	AUC	AUPR	Accuracy	Precision	Recall	F1-score
B-dataset	W/O-GM	0.8764	0.8809	0.8126	0.8188	0.8674	0.8496
	W/O-GPS	0.9083	0.9092	0.8332	0.8407	0.8854	0.8755
	W/O-MTA	0.9016	0.9005	0.8223	0.8296	0.8715	0.8621
	W/O-LPE	0.9126	0.9148	0.8438	0.8473	0.9061	0.8810
	W/O-RWPE	0.9147	0.9152	0.8441	0.8460	0.9082	0.8861
	W/O-LR	0.9062	0.9079	0.8365	0.8381	0.8951	0.8795
	MMADPE	0.9209	0.9231	0.8597	0.8533	0.9124	0.8976
F-dataset	W/O-GM	0.9203	0.9177	0.8606	0.8541	0.9007	0.8827
	W/O-GPS	0.9516	0.9525	0.8974	0.8750	0.9232	0.8903
	W/O-MTA	0.9550	0.9551	0.9001	0.8804	0.9295	0.8977
	W/O-LPE	0.9572	0.9617	0.8970	0.8786	0.9218	0.8995
	W/O-RWPE	0.9569	0.9569	0.8965	0.8728	0.9291	0.8998
	W/O-LR	0.9509	0.9537	0.8848	0.8617	0.9164	0.8983
	MMADPE	0.9611	0.9613	0.9094	0.8903	0.9391	0.9078
C-dataset	W/O-GM	0.9348	0.9369	0.8743	0.8702	0.9122	0.8935
	W/O-GPS	0.9600	0.9597	0.9030	0.8976	0.9353	0.9104
	W/O-MTA	0.9546	0.9571	0.8977	0.8841	0.9296	0.9015
	W/O-LPE	0.9613	0.9661	0.9024	0.8890	0.9202	0.9041
	W/O-RWPE	0.9624	0.9651	0.9039	0.8867	0.9260	0.9057
	W/O-LR	0.9544	0.9549	0.8963	0.8797	0.9152	0.9001
	MMADPE	0.9682	0.9725	0.9160	0.8975	0.9436	0.9176

Bold identifies the maximum value within each column

indicating that introducing proteins as mediators provides the model with crucial multi-hop topological evidence, thereby enhancing its ability to retrieve sparse positive examples. More importantly, compared to the representative baseline method AMDGT, which also constructs a tripartite heterogeneous network, MMADPE shows consistent improvements across multiple metrics, such as AUC, AUPR, precision, and accuracy. These stable enhancements not only reflect an increase in the model's overall discriminative ability and ranking quality but also demonstrate its stronger modeling capability in discovering higher-order heterogeneous interactions achieved through mediating nodes.

Ablation studies

To validate the effectiveness of the core modules, we conducted ablation studies on five key components: W/O-GM (without Graph Mamba), W/O-GPS (without GraphGPS), W/O-MTA (without multi-hop aggregation), W/O-LPE (without Laplacian positional encoding), W/O-RWPE (without random walk positional encoding), and W/O-LR (removing both types of positional encodings).

As illustrated in Fig. 7 and Table 5, experimental results on three datasets demonstrate that removing the Graph Mamba module leads to the most significant performance degradation, confirming its crucial role in capturing long-range dependencies through state-space modeling and edge

Table 6 The top 10 potential drugs predicted by MMADPE for Stomach Neoplasms

Disease	Drug	DrugBankID	PMID
Stomach Neoplasms	Cytarabine	DB00987	30670679
	Vinblastine	DB00570	25279216
	Carmustine	DB00262	34973135
	Bleomycin	DB00290	34973135
	Daunorubicin	DB00694	26098866
	Vincristine	DB00541	31945315
	Imatinib	DB00619	34973135
	Etoposide	DB00773	35246762
	Dacarbazine	DB00851	23593346
	Cisplatin	DB00515	36914835

gating mechanisms. The performance drop observed in W/O-GPS validates the necessity of GraphGPS for multi-scale feature fusion, where the absence of the GateGCN branch impairs local noise filtering and the lack of the self-attention mechanism compromises global semantic understanding. The performance decline in W/O-MTA substantiates the unique advantage of multi-hop aggregation in modeling biological cascade effects. Similarly, removing positional encodings as seen in W/O-LPE and W/O-RWPE highlights their importance in preserving structural prior

knowledge. Notably, the W/O-LR variant that removes both positional encodings exhibited a more pronounced performance degradation (AUROC decreased by 1.42% and AUPR decreased by 1.76%), further confirming that Laplacian positional encoding and random walk positional encoding provide complementary contributions to representing graph structural information. Among all variants, W/O-GM shows the poorest performance, indicating that the global–local synergy within Graph Mamba serves as the primary performance driver. Meanwhile, multi-hop aggregation, Graph-GPS, and positional encodings provide complementary support from perspectives of cascade effect modeling, multi-scale fusion, and structural awareness, respectively. In conclusion, these five core modules collectively contribute to MMADPE’s superior performance through their synergistic interactions.

Case study

To evaluate the clinical application value of our model, we conducted experiments in the fields of Stomach Neoplasms and Type 2 Diabetes Mellitus (T2DM) to assess relevant drug–disease associations. In this study, we utilized all known association data from C-Dataset for model training. For unobserved associations related to Stomach Neoplasms or T2DM, we predicted these associations through scoring and ranked the candidate drug–disease pairs in descending order based on prediction probability scores, prioritizing higher-ranked pairs as the most likely potential associations.

Stomach Neoplasms are highly prevalent malignant tumors of the digestive tract, with incidence closely linked to dietary habits. Early-stage symptoms are often subtle, while advanced-stage treatment poses significant challenges, seriously threatening patients’ health and survival. As shown in Table 6, all top 10 predicted drugs have been validated in relevant medical literature. Our MMADPE model predicts

Cytarabine as a potential therapeutic agent for Stomach Neoplasms. Research [56] demonstrates that abnormal ACTC1 expression may promote tumor progression by regulating cytoskeletal remodeling and paracrine signaling. Cytarabine significantly reduces ACTC1 expression, disrupts the protective tumor microenvironment, and enhances chemotherapeutic efficacy against tumor cells [57]. We hypothesize that Cytarabine’s mol2vec embedding encodes its chemical properties as a cytosine analog, allowing the model to link this compound to interventions targeting key pathological processes in gastric cancer. BCL2L1, a key anti-apoptotic protein in the Bcl-2 family, is highly expressed in Stomach Neoplasms, while CASP8, a crucial initiator of extrinsic apoptosis pathways, shows inactivation or low expression correlated with disease progression. The DNA alkylating agent Carmustine not only directly damages tumor cells but may also enhance pro-apoptotic effects by downregulating BCL2L1 and activating CASP8. The model’s prediction of Carmustine likely stems from its recognition of the drug’s alkylating properties. Furthermore, Dacarbazine may serve as a potential candidate by targeting CDKN1A-related pathways [58], particularly for p53 wild-type or CDKN1A low expression subtypes.

Type 2 Diabetes Mellitus (T2DM) is a common chronic metabolic disorder characterized by insulin resistance and relative insulin deficiency, strongly associated with obesity and poor lifestyle habits. Prolonged hyperglycemia can lead to various severe complications. Table 7 presents the top 10 most probable drug candidates for T2DM, with 8 already validated in medical literature. MMADPE predicts Busulfan as a potential therapeutic for T2DM, this drug may activate BAX-dependent apoptotic pathways through ROS generation [59], clearing dysfunctional pancreatic islet cells and promoting healthy β -cell regeneration to restore islet function. We speculate that the model inferred Busulfan’s cross-indication potential based on its phenotypic outcomes (e.g., regenerative pathways) and multi-hop associations with β -cell dysfunction. Penicillamine could inhibit the PDK1–AKT signaling pathway through copper ion chelation, thereby regulating IRS1 phosphorylation status and GLUT4 membrane localization (both core targets in diabetic insulin resistance). By blocking copper-dependent PDK1–AKT overactivation, Penicillamine may restore normal insulin signaling [60]. The prediction of Penicillamine effectively demonstrates the value of biochemical semantics, its mol2vec features capture its chelating properties, and the model successfully associates this chemical function (“thiol-based metal binding”) with disrupting metal-dependent signaling pathways (PDK1–AKT). This exemplifies how the model leverages molecular-level embeddings to derive biologically plausible repurposing hypotheses. Additionally, Nilotinib may improve systemic insulin sensitivity and delay diabetic complications by suppressing the CD36–NF- κ B

Table 7 The top 10 potential drugs predicted by MMADPE for Diabetes Mellitus, Type 2

Disease	Drug	DrugBankID	PMID
Diabetes Mellitus, Type 2	Tirofiban	DB00775	N/A
	Busulfan	DB01008	31626838
	Penicillamine	DB00859	24513509
	Nilotinib	DB04868	32715377
	Perindopril	DB00790	31626838
	Docetaxel	DB01248	28182703
	Methotrexate	DB00563	32715377
	Prednisone	DB00635	31626838
	Omacetaxine mepesuccinate	DB04865	N/A
	Mifepristone	DB00834	39059604

Molecular docking experiments

This study focuses on type 2 diabetes mellitus (T2DM) and systematically evaluates the clinical translational potential and underlying mechanistic plausibility of the predictions made by the MMADPE model. To achieve this, molecular docking analyses were conducted to quantitatively assess the interactions between the top 10 candidate drugs predicted by the model and five key target proteins implicated in T2DM pathophysiology, utilizing AutoDock Vina software [62, 63]. The detailed results are presented in Table 8. The findings indicate that the absolute value of the binding energy between drug molecules and target proteins is positively correlated with their potential therapeutic efficacy. In other words, higher absolute binding energy reflects stronger binding affinity, suggesting greater potential for development as anti-diabetic drug candidates. Importantly, the strong binding energies observed for several MMADPE-predicted drugs (e.g., Mifepristone, Nilotinib, Docetaxel) provide orthogonal validation at the molecular interaction level. This suggests that the feature representations learned by MMADPE, which combine chemical structure mol2vec, disease semantics mesh2vec, and network topologies P_{LPE} and P_{RWPE} , are capable of implicitly capturing aspects of molecular complementarity relevant to target binding. It should be noted that the current molecular docking results provide only theoretical predictions of drug–target interactions. Therefore, further functional validation through comprehensive in vitro and in vivo experiments remains necessary to confirm their therapeutic viability.

Following the molecular docking experiments, an in-depth investigation was conducted to further evaluate the binding characteristics of two unvalidated candidate drugs, Tirofiban and Omacetaxine mepesuccinate, with T2DM-associated target proteins. To elucidate their interaction patterns, molecular visualization analyses were performed using PyMOL and Discovery Studio software (Fig. 8). The results revealed that Tirofiban formed distinct hydrogen bonds with the target protein 1X70. Moreover, the interactions involving Tirofiban and Omacetaxine mepesuccinate included not only hydrogen bonding but also hydrophobic interactions, such as π -alkyl interactions, carbon–hydrogen bonds, and alkyl interactions. These specific interaction patterns revealed by docking offer potential mechanistic hypotheses for how these MMADPE-predicted drugs might exert effects in T2DM. For instance, Tirofiban's interaction with 1X70 suggests a potential modulation mechanism distinct from its primary antiplatelet action. While the model did not predict specific targets, the high prediction score combined with favorable docking energies and specific interaction motifs lends credence to the biological plausibility of these repositioning suggestions derived from MMADPE's integrative analysis.

Overall, these findings highlight the potential value and biological coherence of the T2DM candidate drugs predicted by MMADPE. The mutual validation between the prediction results and molecular docking experiments further strengthens the necessity for experimental investigation of these candidate drugs.

Conclusion

In this study, we propose a novel computational model named MMADPE for predicting drug–disease associations. The model employs a dual-channel approach to learn high-order feature representations of drugs and diseases. On one hand, it utilizes a multi-hop aggregation-based Graph Mamba architecture to extract similarity interaction features from homogeneous networks, while on the other, it integrates dual-modal graph positional encoding with a Graph GPS hybrid architecture to capture network interaction features from heterogeneous networks. The fused features are then fed into a dot product decoder for prediction. Experimental results demonstrate that MMADPE outperforms seven state-of-the-art baseline methods in both predictive accuracy and robustness. Furthermore, case studies on two diseases and molecular docking experiments provide additional validation of its reliability and practical application value.

Future research will focus on the following two aspects. First, the current association network lacks richness in biochemical information, and it is necessary to incorporate metabolite data to enhance drug–disease associations and provide more expressive feature information. Second, the existing high-order feature fusion relies only on simple concatenation, which, although preserves multi-view information, lacks adaptive fusion capability. Therefore, more effective fusion strategies need to be explored.

Author contributions PL initiated the project, designed the computational framework, and drafted the manuscript. ML optimized algorithms, validated results, and contributed to manuscript preparation. FG interpreted the biomedical significance and provided critical revisions. All authors read and approved the final manuscript.

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Data availability The data is freely available at GitHub: https://github.com/LUTGraphGroup/MMADPE_LMX.

Code availability The code is freely available at GitHub: https://github.com/LUTGraphGroup/MMADPE_LMX.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Rifaioğlu AS, Atas H, Martin MJ, Cetin-Atalay R, Atalay V, Doğan T (2019) Recent applications of deep learning and machine intelligence on in silico drug discovery: methods, tools and databases. *Brief Bioinform* 20(5):1878–1912. <https://doi.org/10.1093/bib/bby061>
- Cai L, Lu C, Xu J, Meng Y, Wang P, Fu X, Zeng X, Su Y (2021) Drug repositioning based on the heterogeneous information fusion graph convolutional network. *Brief Bioinform* 22(6):319. <https://doi.org/10.1093/bib/bbab319>
- Ciociola AA, Cohen LB, Kulkarni P, Kefalas C, Buchman A, Burke C, Cain T, Connor J, Ehrenpreis ED, Fang J (2014) How drugs are developed and approved by the fda: current process and future directions. *Official journal of the American College of Gastroenterology* ACG 109(5), 620–623. <https://doi.org/10.1038/ajg.2013.407>
- Krantz A (1998) Diversification of the drug discovery process. *Nat Biotechnol* 16(13):1294–1294. <https://doi.org/10.1038/4243>
- Avorn J (2015) The \$2.6 billion pill methodologic and policy considerations. *New England Journal of Medicine* 372(20), 1877–1879. <https://doi.org/10.1056/nejmp1500848>
- Ezzat A, Wu M, Li X-L, Kwok C-K (2019) Computational prediction of drug-target interactions using chemogenomic approaches: an empirical survey. *Brief Bioinform* 20(4):1337–1357. <https://doi.org/10.1093/bib/bby002>
- Raslan MA, Raslan SA, Shehata EM, Mahmoud AS, Sabri NA (2023) Advances in the applications of bioinformatics and chemoinformatics. *Pharmaceuticals* 16(7):1050. <https://doi.org/10.3390/ph16071050>
- Chao W, Quan Z (2021) A machine learning method for differentiating and predicting human-infective coronavirus based on physicochemical features and composition of the spike protein. *Chin J Electron* 30(5):815–823. <https://doi.org/10.1049/cje.2021.06.003>
- Ding Y, Guo F, Tiwari P, Zou Q (2023) Identification of drug-side effect association via multi-view semi-supervised sparse model. *IEEE Transactions on Artificial Intelligence*. <https://doi.org/10.1109/tai.2023.3314405>
- Ding Y, Tiwari P, Zou Q, Guo F, Pandey HM (2022) C-loss based higher order fuzzy inference systems for identifying dna n4-methylcytosine sites. *IEEE Trans Fuzzy Syst* 30(11):4754–4765. <https://doi.org/10.1109/tfuzz.2022.3159103>
- Luo Y, Zhao X, Zhou J, Yang J, Zhang Y, Kuang W, Peng J, Chen L, Zeng J (2017) A network integration approach for drug-target interaction prediction and computational drug repositioning from heterogeneous information. *Nat Commun* 8(1):573. <https://doi.org/10.1038/s41467-017-00680-8>
- Luo H, Li M, Wang S, Liu Q, Li Y, Wang J (2018) Computational drug repositioning using low-rank matrix approximation and randomized algorithms. *Bioinformatics* 34(11):1904–1912. <https://doi.org/10.1093/bioinformatics/bty013>
- Recht B (2011) A simpler approach to matrix completion. *Journal of Machine Learning Research* 12(12). <https://doi.org/10.48550/arXiv.0910.0651>
- Candes EJ, Plan Y (2010) Matrix completion with noise. *Proc IEEE* 98(6):925–936. <https://doi.org/10.1109/jproc.2009.2035722>
- Ding Y, Tang J, Guo F (2020) Identification of drug-target interactions via fuzzy bipartite local model. *Neural Comput Appl* 32(14):10303–10319. <https://doi.org/10.1007/s00521-019-04569-z>
- Sun M, Tiwari P, Qian Y, Ding Y, Zou Q (2022) Mlpsvm-lbs: predicting dna-binding proteins via a multiple laplacian regularized support vector machine with local behavior similarity. *Knowl-Based Syst* 250:109174. <https://doi.org/10.1016/j.knsys.2022.109174>
- LaValley MP (2008) Logistic regression. *Circulation* 117(18):2395–2399. <https://doi.org/10.1161/CIRCULATIONAHA.106.682658>
- Breiman L (2001) Random forests machine learning 45:5–32. <https://doi.org/10.1023/a:1010933404324>
- Gottlieb A, Stein GY, Ruppin E, Sharan R (2011) Predict: a method for inferring novel drug indications with application to personalized medicine. *Mol Syst Biol* 7(1):496. <https://doi.org/10.1038/msb.2011.26>
- Yang K, Zhao X, Waxman D, Zhao X-M (2019) Predicting drug-disease associations with heterogeneous network embedding. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29(12). <https://doi.org/10.1063/1.5121900>
- Kitsiranuwat S, Suratanee A, Plaimas K (2022) Integration of various protein similarities using random forest technique to infer augmented drug-protein matrix for enhancing drug-disease association prediction. *Sci Prog* 105(3):00368504221109215. <https://doi.org/10.1177/00368504221109215>
- Zeng X, Zhu S, Liu X, Zhou Y, Nussinov R, Cheng F (2019) Deepdr: a network-based deep learning approach to in silico drug repositioning. *Bioinformatics* 35(24):5191–5198. <https://doi.org/10.1093/bioinformatics/btz418>
- Liu H, Zhang W, Song Y, Deng L, Zhou S (2020) Hnet-dnn: inferring new drug-disease associations with deep neural network based on heterogeneous network features. *J Chem Inf Model* 60(4):2367–2376. <https://doi.org/10.1021/acs.jcim.9b01008>
- Meng Y, Lu C, Jin M, Xu J, Zeng X, Yang J (2022) A weighted bilinear neural collaborative filtering approach for drug repositioning. *Brief Bioinform* 23(2):581. <https://doi.org/10.1093/bib/bbab581>
- Yang R, Fu Y, Zhang Q, Zhang L (2024) Gcngat: drug-disease association prediction based on graph convolution neural network and graph attention network. *Artif Intell Med* 150:102805. <https://doi.org/10.1016/j.artmed.2024.102805>
- Zhao B-W, Hu L, You Z-H, Wang L, Su X-R (2022) Hingrl: predicting drug-disease associations with graph representation learning on heterogeneous information networks. *Brief Bioinform* 23(1):515. <https://doi.org/10.1093/bib/bbab515>
- Zhao B-W, Su X-R, Hu P-W, Ma Y-P, Zhou X, Hu L (2022) A geometric deep learning framework for drug repositioning over heterogeneous information networks. *Brief Bioinform* 23(6):384. <https://doi.org/10.1093/bib/bbac384>
- Liu J, Guan S, Zou Q, Wu H, Tiwari P, Ding Y (2024) Amdgt: attention aware multi-modal fusion using a dual graph transformer for drug-disease associations prediction. *Knowl-Based Syst* 284:111329. <https://doi.org/10.1016/j.knsys.2023.111329>
- Davis AP, Grondin CJ, Johnson RJ, Sciaky D, King BL, McMoran R, Wiegers J, Wiegers TC, Mattingly CJ (2017) The comparative toxicogenomics database: update 2017. *Nucleic Acids Res* 45(D1):972–978. <https://doi.org/10.1093/nar/gkw838>
- Zhang W, Yue X, Lin W, Wu W, Liu R, Huang F, Liu F (2018) Predicting drug-disease associations by using similarity constrained

- matrix factorization. *BMC Bioinformatics* 19(1):233. <https://doi.org/10.1186/s12859-018-2220-4>
31. Luo H, Wang J, Li M, Luo J, Peng X, Wu F-X, Pan Y (2016) Drug repositioning based on comprehensive similarity measures and bi-random walk algorithm. *Bioinformatics* 32(17):2664–2671. <https://doi.org/10.1093/bioinformatics/btw228>
 32. Wishart DS, Feunang YD, Guo AC, Lo EJ, Marcu A, Grant JR, Sajed T, Johnson D, Li C, Sayeeda Z (2018) Drugbank 5.0: a major update to the drugbank database for 2018. *Nucleic acids Res* 46(D1):1074–1082. <https://doi.org/10.1093/nar/gkx1037>
 33. Hamosh A, Scott AF, Amberger JS, Bocchini CA, McKusick VA (2005) Online mendelian inheritance in man (omim), a knowledgebase of human genes and genetic disorders. *Nucleic acids Res* 33(suppl_1):514–517. <https://doi.org/10.1093/nar/gki033>
 34. Yang H, Ding Y, Tang J, Guo F (2021) Drug-disease associations prediction via multiple kernel-based dual graph regularized least squares. *Appl Soft Comput* 112:107811. <https://doi.org/10.1016/j.asoc.2021.107811>
 35. Guha R (2007) Chemical informatics functionality in r. *Journal of Statistical Software* 18, 1–16. <https://doi.org/10.18637/jss.v018.i05>
 36. Van Driel MA, Bruggeman J, Vriend G, Brunner HG, Leunissen JA (2006) A text-mining analysis of the human phenome. *Eur J Hum Genet* 14(5):535–542. <https://doi.org/10.1038/sj.ejhg.5201585>
 37. Jaeger S, Fulle S, Turk S (2018) Mol2vec: unsupervised machine learning approach with chemical intuition. *J Chem Inf Model* 58(1):27–35. <https://doi.org/10.1021/acs.jcim.7b00616>
 38. Guo Z-H, You Z-H, Huang D-S, Yi H-C, Zheng K, Chen Z-H, Wang Y-B (2021) Meshheading2vec: a new method for representing mesh headings as vectors based on graph embedding algorithm. *Brief Bioinform* 22(2):2085–2095. <https://doi.org/10.1093/bib/bbaa037>
 39. Baumann N (2016) How to use the medical subject headings (mesh). *Int J Clin Pract* 70(2):171–174. <https://doi.org/10.1111/ijcp.12767>
 40. Liu W, Lu P, Gao J (2025) Rgcgt: a high-order feature learning framework for predicting disease-metabolite interaction using residual graph convolution and graph transformer. *Expert Syst Appl*. <https://doi.org/10.1016/j.eswa.2025.126637>
 41. Abdi H, Williams LJ (2010) Principal component analysis. *Wiley Interdiscip Rev Comput Stat* 2(4):433–459. <https://doi.org/10.1002/wics.101>
 42. Sun Z, Wang C, Hu W, Chen M, Dai J, Zhang W, Qu Y (2020) Knowledge graph alignment network with gated multi-hop neighborhood aggregation. *IProceedings of the AAAI Conference on Artificial Intelligence*. 34:222–229. <https://doi.org/10.1609/aaai.v34i01.5354>
 43. Scarselli F, Gori M, Tsoi AC, Hagenbuchner M, Monfardini G (2008) The graph neural network model. *IEEE Trans Neural Netw* 20(1):61–80. <https://doi.org/10.1109/TNN.2008.2005605>
 44. Yun S, Jeong M, Kim R, Kang J, Kim HJ (2019) Graph transformer networks. *Adv Neural Inf Process Syst*. 32. <https://doi.org/10.48550/arXiv.1911.06455>
 45. Wang C, Tsepa O, Ma J, Wang B (2024) Graph-mamba: Towards long-range graph sequence modeling with selective state spaces. *arXiv preprint arXiv:2402.00789*. <https://doi.org/10.48550/arXiv.2402.00789>
 46. Kreuzer D, Beaini D, Hamilton W, Létourneau V, Tossou P (2021) Rethinking graph transformers with spectral attention. *Adv Neural Inform Process Syst* 34:21618–21629. <https://doi.org/10.48550/arXiv.2106.03893>
 47. Dwivedi VP, Luu AT, Laurent T, Bengio Y, Bresson X (2021) Graph neural networks with learnable structural and positional representations. *arXiv preprint arXiv:2110.07875*. <https://doi.org/10.48550/arXiv.2110.07875>
 48. Rampášek L, Galkin M, Dwivedi VP, Luu AT, Wolf G, Beaini D (2022) Recipe for a general, powerful, scalable graph transformer. *Adv Neural Inform Process Syst* 35:14501–14515. <https://doi.org/10.48550/arXiv.2205.12454>
 49. Kingma DP, Ba J (2014) Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*. <https://doi.org/10.48550/arXiv.1412.6980>
 50. Ngo NK, Hy TS, Kondor R (2022) Predicting drug-drug interactions using deep generative models on graphs. *arXiv preprint arXiv:2209.09941*. <https://doi.org/10.48550/arXiv.2209.09941>
 51. Min E, Chen R, Bian Y, Xu T, Zhao K, Huang W, Zhao P, Huang J, Ananiadou S, Rong Y (2022) Transformer for graphs: an overview from architecture perspective. *arXiv preprint arXiv:2202.08455*. <https://doi.org/10.48550/arXiv.2202.08455>
 52. Wang X, Ji H, Shi C, Wang B, Ye Y, Cui P, Yu PS (2019) Heterogeneous graph attention network. In: *The World Wide Web Conference*, pp. 2022–2032. <https://doi.org/10.1145/3308558.3313562>
 53. Ma M, Lei X (2023) A dual graph neural network for drug-drug interactions prediction based on molecular structure and interactions. *PLoS Comput Biol* 19(1):1010812. <https://doi.org/10.1371/journal.pcbi.1010812>
 54. Piñero J, Bravo, À, Queralt-Rosinach N, Gutiérrez-Sacristán A, Deu-Pons J, Centeno E, García-García J, Sanz F, Furlong LI (2016) Disgenet: a comprehensive platform integrating information on human disease-associated genes and variants. *Nucleic Acids Res*. 943. <https://doi.org/10.1093/nar/gkw943>
 55. Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, Smetanin N, Verkuil R, Kabeli O, Shmueli Y (2023) Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 379(6637):1123–1130. <https://doi.org/10.1126/science.ade2574>
 56. Somaiah C, Kumar A, Sharma R, Sharma A, Anand T, Bhattacharyya J, Das D, Deka Talukdar S, Jaganathan BG (2018) Mesenchymal stem cells show functional defect and decreased anti-cancer effect after exposure to chemotherapeutic drugs. *J Biomed Sci* 25:1–10. <https://doi.org/10.1186/s12929-018-0407-7>
 57. Paul-Samojedny M, Pudełko A, Kowalczyk M, Fila-Daniłow A, Suchanek-Raif R, Borkowska P, Kowalski J (2016) Combination therapy with akt3 and pi3kca sirna enhances the antitumor effect of temozolomide and carmustine in t98g glioblastoma multiforme cells. *BioDrugs* 30:129–144. <https://doi.org/10.1007/s40259-016-0160-y>
 58. Esimbekova AR, Palkina NV, Zinchenko IS, Belenyuk VD, Savchenko AA, Sergeeva EY, Ruksha TG (2023) Focal adhesion alterations in g0-positive melanoma cells. *Cancer Med* 12(6):7294–7308. <https://doi.org/10.1002/cam4.5510>
 59. Nasimi P, Tabandeh MR, Vahdati A, Khatamsaz S (2015) Busulfan induces oxidative stress-and bcl-2 family gene-related apoptosis in epididymal sperm and testis of adult male mice. *Physiol Pharmacol* 19(3):208–215. <https://doi.org/10.12692/ijb/6.5.70-78>
 60. Guo J, Cheng J, Zheng N, Zhang X, Dai X, Zhang L, Hu C, Wu X, Jiang Q, Wu D (2021) Copper promotes tumorigenesis by activating the pdk1-akt oncogenic pathway in a copper transporter 1 dependent manner. *Adv Sci* 8(18):2004303. <https://doi.org/10.1002/advs.202004303>
 61. Trojani A, Pungolino E, Rossi G, D Adda M, Lodola M, Camillo BD, Perego A, Turrini M, Orlandi E, Borin L (2018) Wide-transcriptome analysis and cellularity of bone marrow cd34+/

- lin-cells of patients with chronic-phase chronic myeloid leukemia at diagnosis vs. 12 months of first-line nilotinib treatment. *Cancer Biomarkers* 21(1):41–53. <https://doi.org/10.3233/cbm-170209>
62. Morris GM, Huey R, Lindstrom W, Sanner MF, Belew RK, Goodsell DS, Olson AJ (2009) Autodock4 and autodocktools4: automated docking with selective receptor flexibility. *J Comput Chem* 30(16):2785–2791. <https://doi.org/10.1002/jcc.21256>
63. Trott O, Olson AJ (2010) Autodock vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J Comput Chem* 31(2):455–461. <https://doi.org/10.1002/jcc.21334>

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